MIL-B-13900D
21 August 1967
SUPERSEDING
MIL-B-0013900C(MU)
5 October 1965
MIL-B-13900A
25 April 1956

MILITARY SPECIFICATION

BODY, IGNITION CYLINDER

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of ignition cylinder body.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

STANDARDS

FEDERAL

Fed. Test Method Std. No. 406 - Plastics: Methods of Testing.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

DRAWINGS

US ARMY EDGEWOOD ARSENAL

C81-1-1289 - Body, Ignition Cylinder.

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(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification Ratings, Rules and Regulations.

(Application for copies of these freight classification rules should be addressed to the Uniform Classification Committee, 202 Union Station, Chicago, Illinois 60606.)

3. REQUIREMENTS

3.1 Materials. All materials used in the manufacture of the body shall conform to the specifications and drawings that form a part of this specification. In addition the body shall be molded of thermo-setting phenolic general purpose grade material which is compatible with adhesives and coatings including silicone base materials. Mold releases employed in fabricating the bodies shall be likewise compatible or easily removable using standard cleaning procedures.

3.2 Manufacture. The body shall be manufactured as specified on Drawing C81-1-1289.

3.3 Physical properties.

3.3.1 Resistance to cold. The body shall not warp, crack or undergo dimensional changes when tested as specified in 4.3.4.1.

3.3.2 Accelerated aging. The body shall not warp, crack or undergo dimensional changes when tested as specified in 4.3.4.2.

3.3.3 Resistance to impact. The body shall not fracture or be permanently distorted when tested as specified in 4.3.4.3.

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3.4 Flammability. Test strips measuring 5 inches long by 0.5 inch wide by 0.005 ± .001 inch thick shall be made from the molding powder used for each preproduction lot. The test strips shall be self-extinguishing and not burn when tested as specified in 4.3.4.4.

3.5 Preproduction. Prior to the start of regular production a preproduction sample of bodies will be produced in accordance with this specification for examination and tests (see 4.2).

3.6 Workmanship. The body shall be free from contamination (foreign matter, oil or grease) and damage such as cracks or deformation which could affect its intended use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Preproduction.

4.2.1 Sample. The supplier shall produce a sufficient quantity of bodies so that five each may be randomly selected from each mold cavity. These bodies shall be produced using the same materials, methods, equipment, processes and procedures as will be used during regular production. Any changes in materials, methods, processes or procedures will require a new preproduction sample.

4.2.2 Inspection. Sampling and inspection shall be in accordance with table I.

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Table I. Preproduction sampling, inspection, acceptance/rejection criteria

Quantity	Requirement	Tests	Bodies	
			Acc	Rej
6 bodies	3.3.1	4.3.4.1	0	1
6 bodies	3.3.2	4.3.4.2	0	1
6 bodies	3.3.3	4.3.4.3	0	1
5 test strips	3.4	4.3.4.4	0	1
All produced	Dwg C81-1-1289	4.3.3.2	0	1

4.3 Regular production.

4.3.1 Lotting. A lot shall consist of the bodies produced from one molding machine by one manufacturer, from the same materials and under essentially the same manufacturing conditions as were used in the preproduction lot. However, no more than one batch (see 6.3) of molding powder shall be represented in any one lot of finished bodies.

4.3.2 Sampling for examination. Sampling shall be conducted in accordance with MIL-STD-105.

4.3.3 Inspection.

4.3.3.1 Examination. Sample bodies shall be examined in accordance with the classification of defects and MIL-STD-105.

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4.3.3.2 Classification of defects.

(a) Body, ignition cylinder (Dwg C81-1-1289).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Diameter (0.945) incorrect
102	Depth (0.943) incorrect
103	Closure plate cavity incorrect
104	Match hole diameter incorrect
105	Cavity diameter incorrect
106	Cavity out of round or excessive taper
107	Match head mix groove width (0.264) incorrect
108	Match head mix groove (0.141) incorrect
109	Diameter (0.687) incorrect
110	Cavity depth (1.010) incorrect
111	Match holes location or rear surface profile incorrect
112	Cavities location incorrect
113	Center hole dimension (0.317) incorrect
114	Machine grooves depth or size (routing) incorrect
115	Match pin ring counter sinks incorrect
116	Body not identified with mold or mold cavity
117	Workmanship (3.6)

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4.3.4 Tests. The following test shall be performed on the preproduction lots. Prior and subsequent to each conditioning phase (cold, accelerated aging) a dimensional evaluation shall be made to determine the effects of the conditioning phase on the dimensions.

4.3.4.1 Resistance to cold. The body shall be cooled to minus 65° ± 5°F temperature by any convenient method and retained at this temperature for a minimum of 2 hours, removed and allowed to return to room temperature for a minimum of 2 hours. Moisture shall not be removed from the surface of the body during test. If cooling solution is used, the body shall not be placed directly in the solution.

4.3.4.2 Accelerated aging. The body shall be heated to a temperature of 100° ± 5°F and 100 percent humidity for a minimum period of 24 hours, followed by 24 hours at a temperature of 160° ± 5°F dry heat (desiccator) and then 24 hours at existing room temperature and humidity.

4.3.4.3 Impact.

(a) As manufactured. The body shall be subjected to a drop of 3 feet ± 1 inch onto a concrete floor 3 times. The body shall be held with axes of the cavities in a horizontal position. Fractures of the 1/16 inch rim of closure plate recess shall not be considered as failures.

(b) Exposure to low temperature. The body after testing as specified in 4.3.4.1 shall be tested as specified in 4.3.4.3(a).

(c) Accelerated aging. The body after testing as specified in 4.3.4.2 shall be tested as specified in 4.3.4.3(a).

4.3.4.4 Flammability. The test strips shall be tested as specified in method 2022 of Fed Test Method Std No. 406 by the benzine drop method of ignition and shall conform to the requirements of 3.4.

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5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging, level C. The bodies shall be cleaned, dried, preserved and packaged in such a manner which will afford adequate protection and safe delivery from the supply source to the first receiving activity for immediate use.

5.2 Packing, level C. The bodies packaged as specified in 5.1 shall be packed in such manner as to insure carrier acceptance and safe delivery to first domestic destination. Containers shall be in accordance with the Uniform Freight Classification rules or regulations of other carriers applicable to the mode of transportation.

5.3 Marking. In addition to any special marking required by the contract or order, unit and intermediate packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The ignition cylinder body is used as a component of the loaded ignition cylinder which is used in the portable flame thrower gun.

6.2 Ordering data. Procurement documents should specify the following:

(a) The title, number and date of this specification.
(b) Preproduction.

(1) Time allowed for supplier submission of samples for Government test and evaluation after award of contract.
(2) Name and address of test facility when testing is performed by the Government.
(3) Packing and shipping instructions when applicable.
(4) Time required for the Government to notify the supplier whether or not to proceed with production.

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6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

Custodians:

Army - MU

User interest:

Navy - MC

Preparing activity:

Army - MU (EA)

Project No. 1040-0061

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